



“DeepSide: A Deep Learning Framework for Drug Side Effect Prediction”



BATCH - 19

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INTERNAL GUIDE

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AGENDA

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ABSTRACT

Developing new medicines is expensive and time-consuming, especially when unexpected side effects show up during testing. These side effects can be harmful and cause the drug to fail, wasting time and money. To help solve this, our project uses a deep learning system called DeepSide to predict possible side effects of drugs before they reach the testing phase.

We used a large public dataset that includes how different drugs affect genes in human cells. By combining this information with the chemical structure of drugs, we trained several AI models. Among them, a special model using SMILES (a text format that describes molecules) worked the best and gave more accurate predictions than older methods.

This system helps researchers identify dangerous side effects early, improving drug safety and saving development costs..

INTRODUCTION

Making new medicines is a long and costly process. One big problem is that some drugs may cause harmful side effects that are not found until very late, sometimes even after people start using them. This can be dangerous and leads to failure of the drug. To avoid this, researchers are now using computers and artificial intelligence to predict side effects early—before testing on humans.

In this project, we created a system called DeepSide. It uses deep learning (a smart AI technique) to make better predictions by combining: Gene expression data (how cells react to the drug), and Chemical structure of the drug .

We used a big dataset called LINCS L1000, which has results from many lab tests on how drugs affect different types of human cells. We also used another dataset, SIDER, to check if our predictions were correct. We tested five deep learning models, and found that one using the SMILES format (a way to describe chemical structures) gave the best results. Our goal is to help scientists find harmful side effects early and make drug development safer and faster.

LITERATURE REVIEW

Title	Authors	Source	Citation	Methodology	Results	Future Enhancements
Deepside: A Deep Learning Framework for Drug Side Effect Prediction	Vaddi Srivallidevi , Sk. Ruhi	Science direct Volume 14, Issue 05, May 2024,	Srivallidevi, V., & Ruhi, (2024). Deepside. Volume 14, Issue 05, May 2024.	Used full LINCS L1000 dataset; evaluated MLPs and CNNs; multi-model input (CS + GEX); SMILES for CNN input; metrics: accuracy, precision, recall, macro-/micro-AUC.	CNN with SMILES showed 13.0% macro-AUC and 3.1% micro-AUC gain; multi-model MLPs (CS + GEX) best among MLPs; CS more informative than GEX; model predicted previously undocumented but literature-supported side effects.	Extend to new datasets; personalize predictions for patient-level efficacy; integrate into clinical decision systems for real-world personalized treatment guidance.

FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

Functional Requirements:

- Accept chemical structure data (e.g., SMILES strings, molecular fingerprints).
- Train deep learning models (MLP, ResMLP, MMNN, MTNN).
- Predict probability scores for multiple side effects per drug
- Fuse multi-modal inputs using concatenation or summation strategies.
- Generate side effect predictions with confidence scores.

Non-Functional Requirements:

- Handle large-scale drug datasets with high-dimensional input features.
- Optimize for both training speed and prediction accuracy.
- Provide a user-friendly interface or API for researchers or developers.
- Batch processing of drug data
- Easy-to-use CLI for researchers

SOFTWARE AND HARDWARE REQUIREMENTS

Software Requiriements :

Component	Specification
OS	Windows 7 Ultimate
Programming Language	Python 3.7
Front-End	Python
Back-End	Django-ORM
Designing	HTML, CSS, JavaScript
Data Base	MySQL(WAMP Server)

Hardware Requirements :

Component	Requirements
Processor	Intel i5(6 th Gen or Later)
RAM	Minimum 4 GB
Hard Disk	20 GB
Key Board	Standard Windows Keyboard
Storage	SSD with at least 50 GB free

UML DIAGRAMS

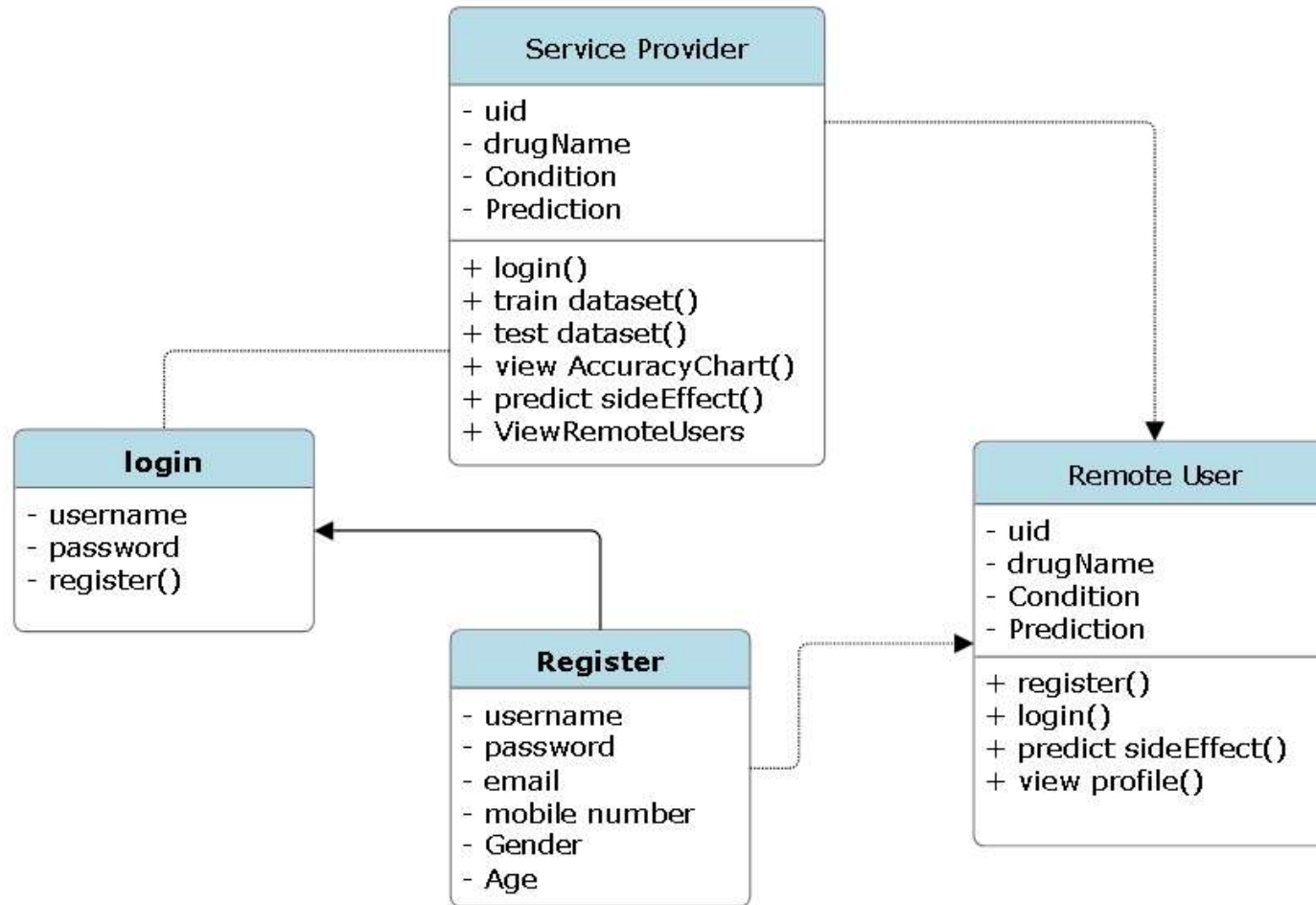
- **Class Diagram**
- **Use Case Diagram**
- **Sequence Diagram**

SYSTEM DESIGN

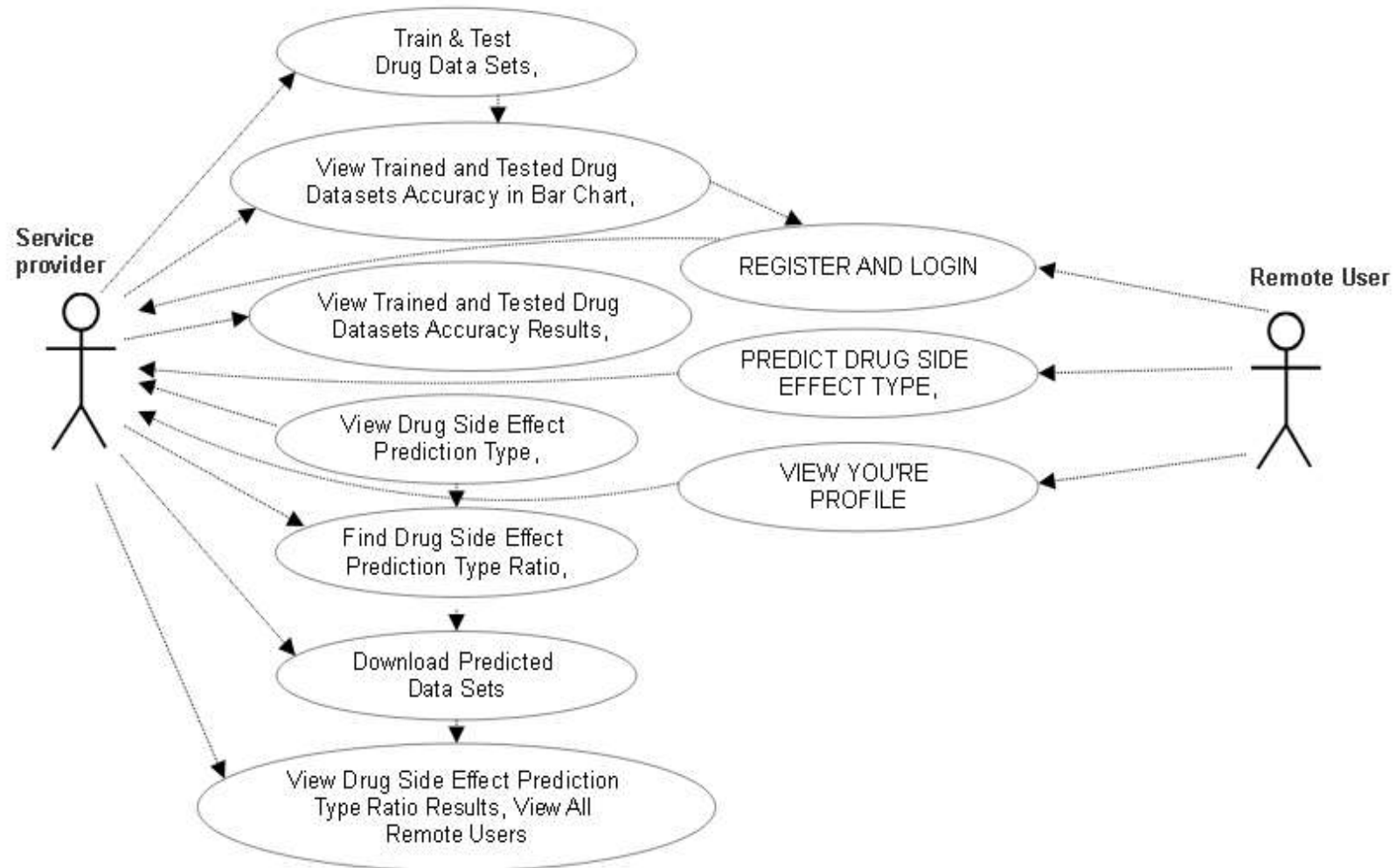
- **Architecture/Blockdiagram**

UML Diagrams

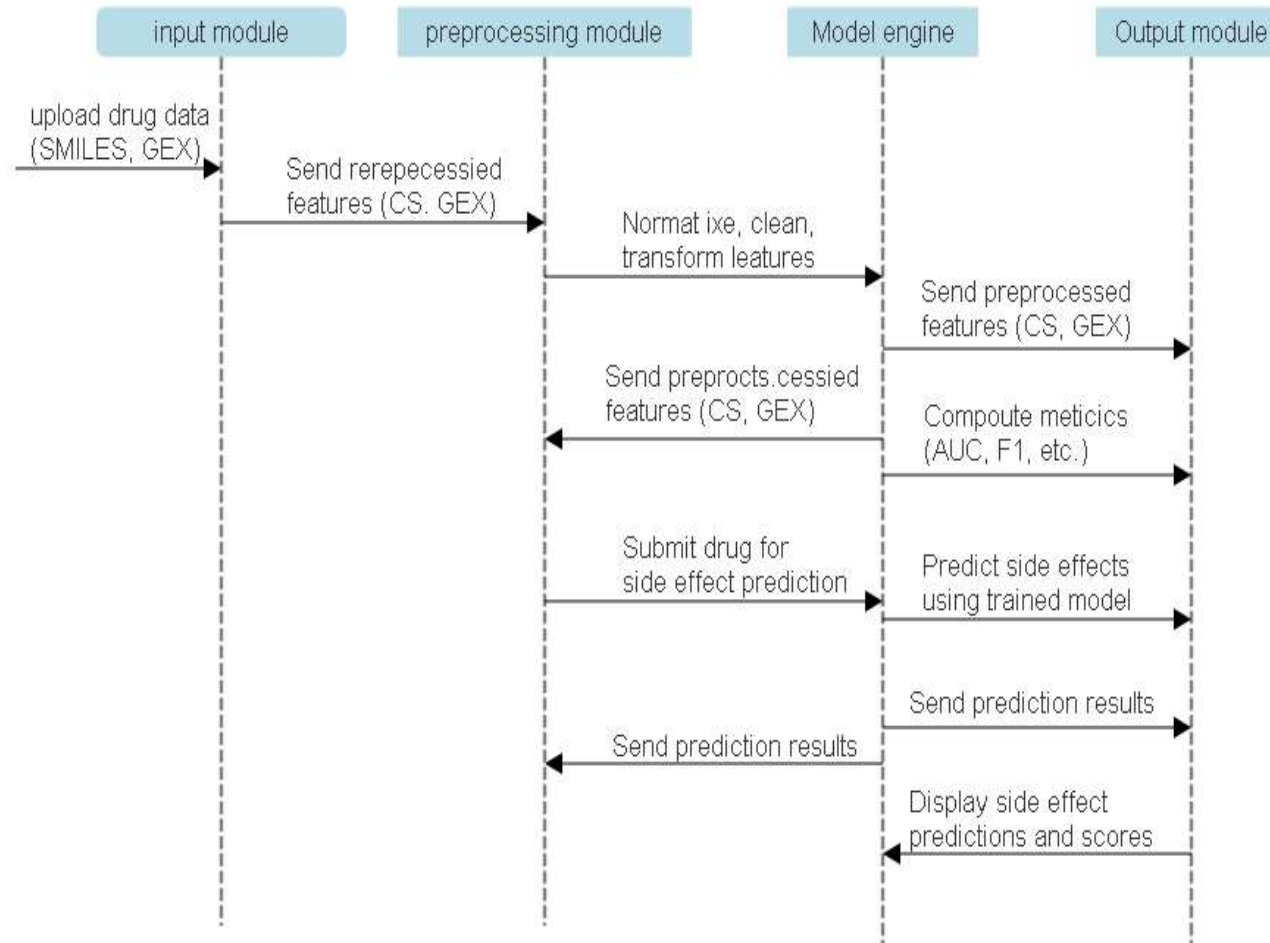
Class Diagram :



Use case:

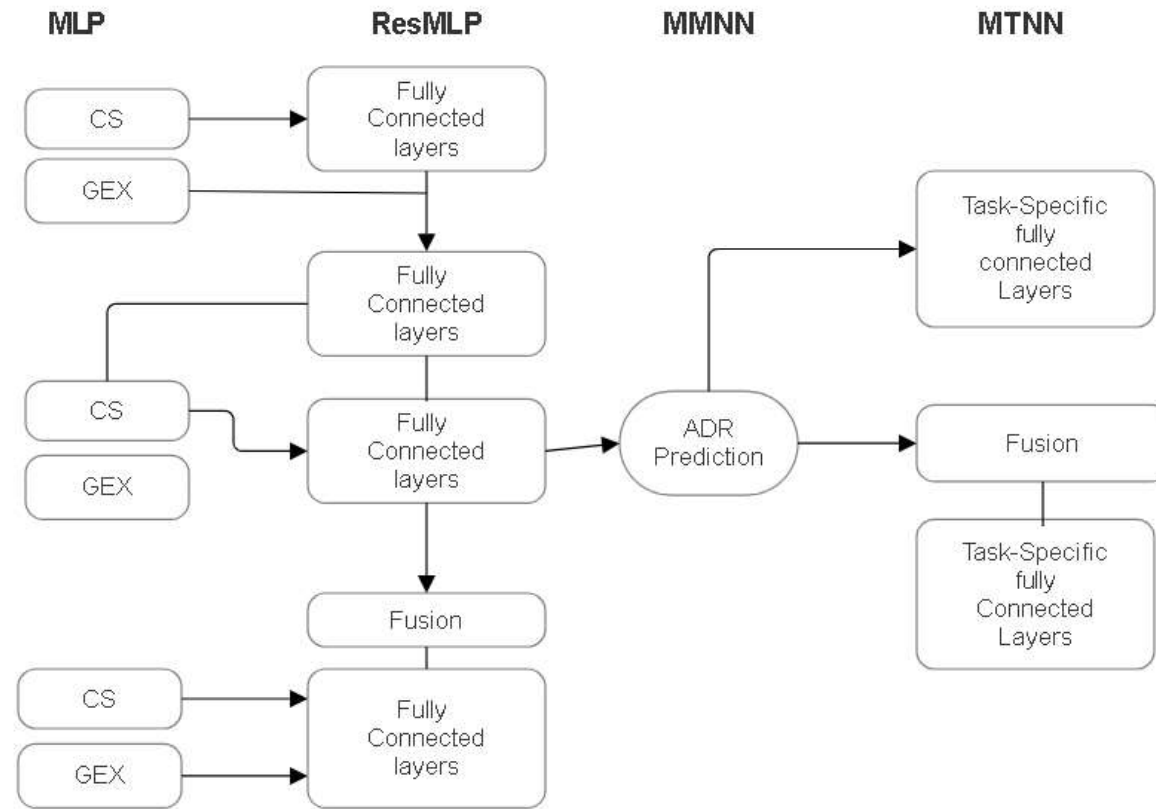


Sequence Diagram:



SYSTEM DESIGN

Architecture/Block diagram:



Implementation :

Views.py :

```
from django.db.models import Count from django.db.models import Q
from django.shortcuts import render, redirect, get_object_or_404
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report from
sklearn.ensemble import VotingClassifier
from Remote_User.models import
ClientRegister_Model,drug_side_effect_prediction,detection_ratio,detection_accuracy from
sklearn.tree import DecisionTreeClassifier
if request.method == "POST" and 'submit1' in request.POST:
    username = request.POST.get('username') password = request.POST.get('password')
    try:
        enter = ClientRegister_Model.objects.get(username=username,password=password)
        request.session["userid"] = enter.id
    return redirect('ViewYourProfile')
```

```
except:
pass

return render(request,'RUser/login.html') def Register1(request):
if request.method == "POST":
username = request.POST.get('username') email = request.POST.get('email') password =
request.POST.get('password') phoneno = request.POST.get('phoneno') country =
request.POST.get('country') state = request.POST.get('state')
city = request.POST.get('city') address = request.POST.get('address') gender =
request.POST.get('gender')
ClientRegister_Model.objects.create(username=username, email=email, password=password,
phoneno=phoneno,
country=country, state=state, city=city, address=address,
gender=gender)
obj = "Registered Successfully"
return render(request, 'RUser/Register1.html', {'object': obj})
```

else:

return render(request,'RUser/Register1.html')

def ViewYourProfile(request): userid = request.session['userid']

obj = ClientRegister_Model.objects.get(id= userid)

return render(request,'RUser/ViewYourProfile.html',{'object':obj})

def Predict_Drug_Side_Effect_Type(request): if request.method == "POST":

uid= request.POST.get('uid')

Drug_Name= request.POST.get('Drug_Name') Condition1= request.POST.get('Condition1')

df = pd.read_csv('Datasets.csv',encoding='latin-1') df

df.columns

def apply_results(Rating): if (int(Rating) <= 7):

return 0 # Low Side Effect else:

return 1 # High Side Effect

df['Results'] = df['Rating'].apply(apply_results)

```
cv = CountVectorizer()
X = df['UID'].apply(str) y = df['Results']
print("UID") print(X) print("Results") print(y)
cv = CountVectorizer()
X = cv.fit_transform(X) models = []
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20)
X_train.shape, X_test.shape, y_train.shape print("Multi-modal neural networks (MMNN)")
from sklearn.neural_network import MLPClassifier mlpc = MLPClassifier().fit(X_train, y_train)
y_pred = mlpc.predict(X_test) print("ACCURACY")
print(accuracy_score(y_test, y_pred) * 100) print("CLASSIFICATION REPORT")
print(classification_report(y_test, y_pred)) print("CONFUSION MATRIX")
print(confusion_matrix(y_test, y_pred))
models.append(('MLPClassifier', mlpc))
```



```
print("SVM")
from sklearn import svm lin_clf = svm.LinearSVC() lin_clf.fit(X_train, y_train)
predict_svm = lin_clf.predict(X_test)
svm_acc = accuracy_score(y_test, predict_svm) * 100 print(svm_acc)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, predict_svm)) print("CONFUSION MATRIX")
print(confusion_matrix(y_test, predict_svm)) models.append(('svm', lin_clf))
print("Logistic Regression")
from sklearn.linear_model import LogisticRegression
reg = LogisticRegression(random_state=0, solver='lbfgs').fit(X_train, y_train) y_pred =
reg.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, y_pred) * 100) print("CLASSIFICATION REPORT")
print(classification_report(y_test, y_pred)) print("CONFUSION MATRIX")
print(confusion_matrix(y_test, y_pred)) models.append(('logistic', reg))
classifier = VotingClassifier(models) classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

```
pred = str(predict_text).replace("[", "") pred1 = pred.replace("]", "")
prediction = int(pred1) if prediction == 0:
val = 'Low Side Effect Found' elif prediction == 1:
val = 'High Side Effect Found'
print(val) print(pred1)
drug_side_effect_prediction.objects.create( uid=uid,
Drug_Name=Drug_Name, Condition1=Condition1,
Prediction=val)
return render(request, 'RUser/Predict_Drug_Side_Effect_Type.html',{'objs': val})
return render(request, 'RUser/Predict_Drug_Side_Effect_Type.html')
```

Output Screens :

Registration page:

DeepSide A Deep Learning Framework for Drug Side Effect Prediction

REGISTER NOW!

REGISTER YOUR DETAILS HERE !!

Enter Username	User Name	Enter Password	Password
Enter EMail Id	Enter Email	Enter Address	Enter Address
Enter Gender	----Select Gender----	Enter Mobile Number	Enter Mobile Number
Enter Country Name	Enter Country Name	Enter State Name	Enter State Name
Enter City Name	Enter City Name		

Register

Login page:

DeepSide A Deep Learning Framework for Drug Side Effect Prediction

Login Using Your Account:

User Name

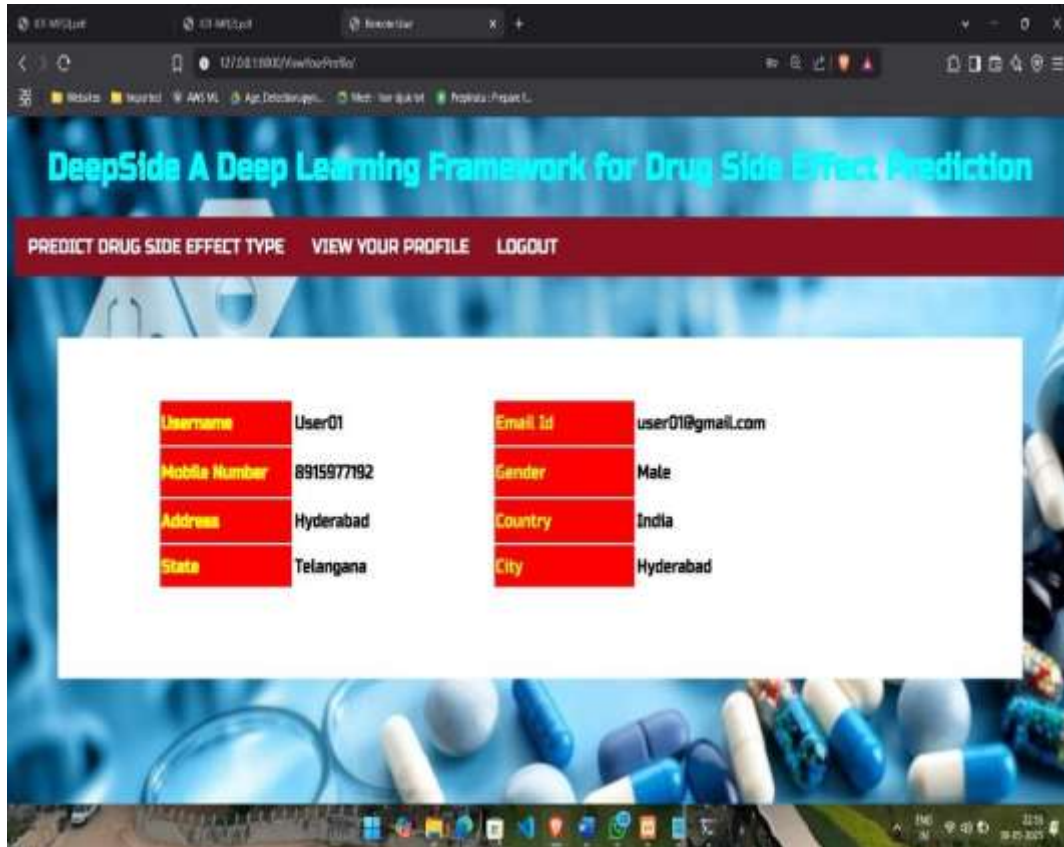
Password

sign_in

Login Using Your Account:

SERVICE PROVIDER **REGISTER**

User Information:



Username	User01	Email Id	user01@gmail.com
Mobile Number	8915977192	Gender	Male
Address	Hyderabad	Country	India
State	Telangana	City	Hyderabad

Predict Drug Side Effect Type:



PREDICTION OF DRUG SIDE EFFECT TYPE !!!

ENTER DRUG DATA SET DETAILS HERE !!!

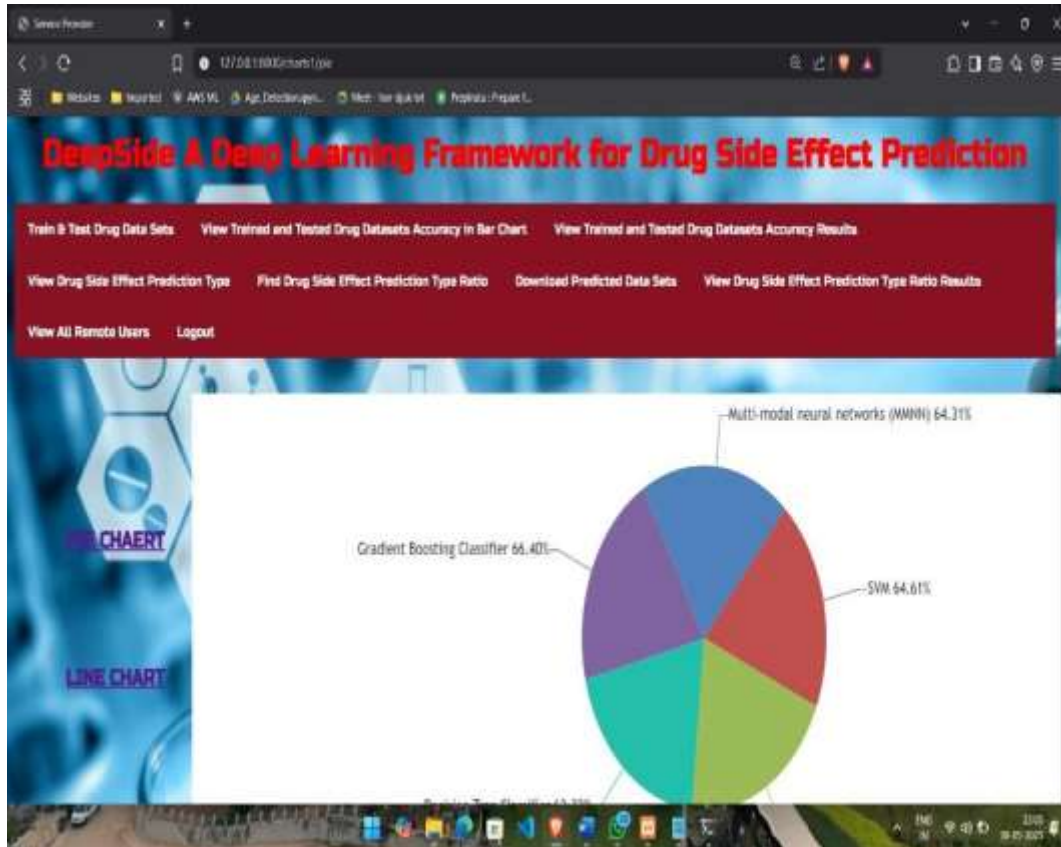
Enter Uid: vho7Qjcx Enter Drug Name: Valsartan

Enter Condition with other Drugs: Left Ventricular Dysfunction

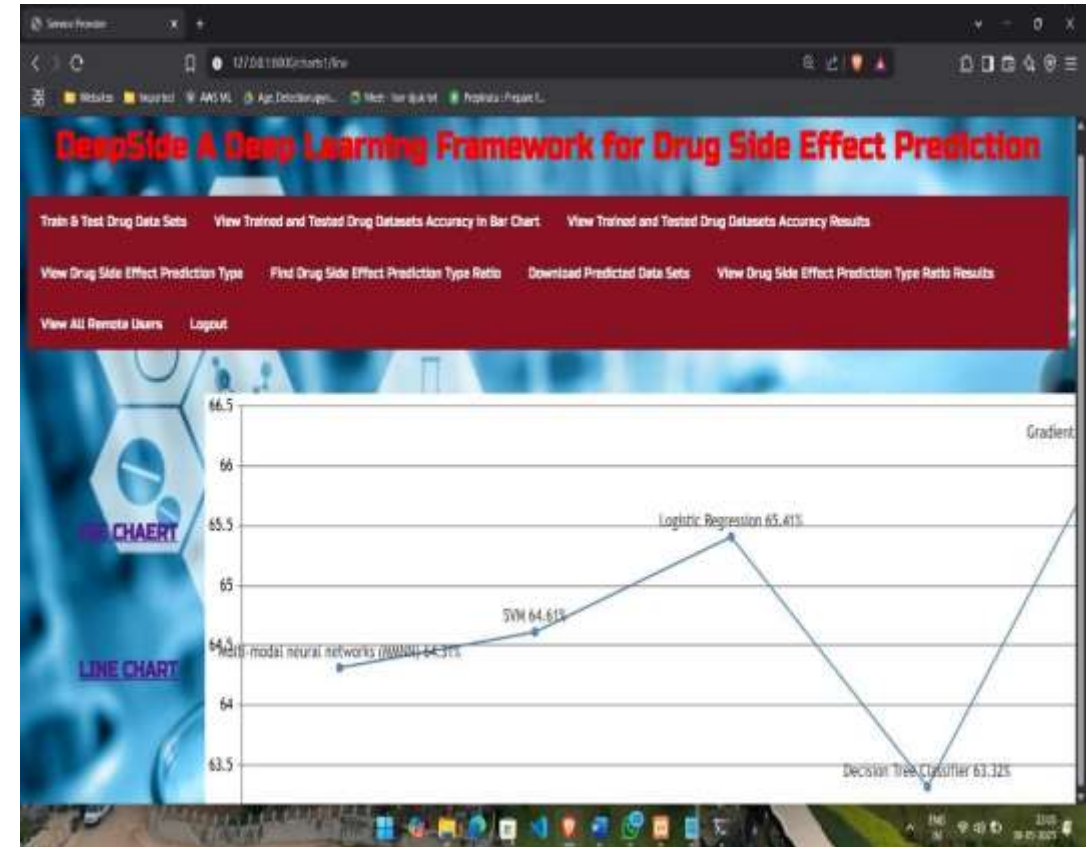
Predict

Predicted Drug Side Effect Type :: High Side Effect Found

Pie Chart:



Line Chart:



CONCLUSION

The pharmaceutical drug development process is a long and demanding process. Unforeseen ADRs that arise at the drug development process can suspend or restart the whole development pipeline. Therefore, the a priori prediction of the side effects of the drug at the design phase is critical. In our Deep Side framework, we use context-related (gene expression) features along with the chemical structure to predict ADRs to account for conditions such as dosing, time interval, and cell line. The proposed MMNN model uses GEX and CS as combined features and achieves better accuracy performance compared to the models that only use the chemical structure (CS) fingerprints. The reported accuracy is noteworthy considering that we are only trying to estimate the condition-independent side effects. Finally, SMILES Conv model outperforms all other approaches by applying convolution on SMILES representation of drug chemical structure.

THANK YOU !